

Prabhakar Lal - React Assignment

Part A - Warm-up: quick concepts

1. In your own words: what is a component in React?

A component is a reusable piece of UI in React. It is usually a JavaScript function that returns JSX.

2. This JSX is broken: `<div class="card">` - what is wrong, and what is the fix?

In JSX, we use `className` instead of `class`.

Correct version: `<div className="card">`

3. Props vs state: which one can a component change itself, and which one is read-only?

State can be changed by the component itself. Props are read-only values passed from a parent component.

4. What does the empty array do in `useEffect(() => { ... }, [])`?

The empty array makes the effect run only once when the component first loads.

5. Why does React need a key prop when you render a list with `.map()`?

React uses the key to identify each list item. This helps React update the list correctly and prevents warnings.

Part B - Bug hunt

BUG 1

Buggy line:

```
return <button onClick={handleSave()}>Save</button>;
```

Corrected line:

```
return <button onClick={handleSave}>Save</button>;
```

Reason:

`handleSave()` calls the function immediately. `handleSave` passes the function to run later when the button is clicked.

BUG 2

Buggy lines:

```
todos.push('New task');  
setTodos(todos);
```

Corrected line:

```
setTodos([...todos, 'New task']);
```

Reason:

State should not be mutated directly. `.push()` changes the original array, so React may not re-render.

BUG 3**Buggy line:**

```
<li>{skill.name}</li>
```

Corrected line:

```
<li key={skill.id}>{skill.name}</li>
```

Reason:

Each item rendered with `.map()` needs a unique key prop.

Part C · Build: "My Mini Profile"

Deployed Link : <https://react-assignment-1-beige.vercel.app/>

Github Link : <https://github.com/Prabhakar0tenn/react-assignment-1>